

1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page. Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.



The image shows the login page of an Airtel router. At the top left is the Airtel logo, and at the top right is a language dropdown menu set to 'English'. Below these are two input fields: 'Username' and 'Password'. At the bottom is a large grey button labeled 'Sign in'.

2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

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Windows IP Configuration

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
Wireless LAN adapter Local Area Connection* 10:

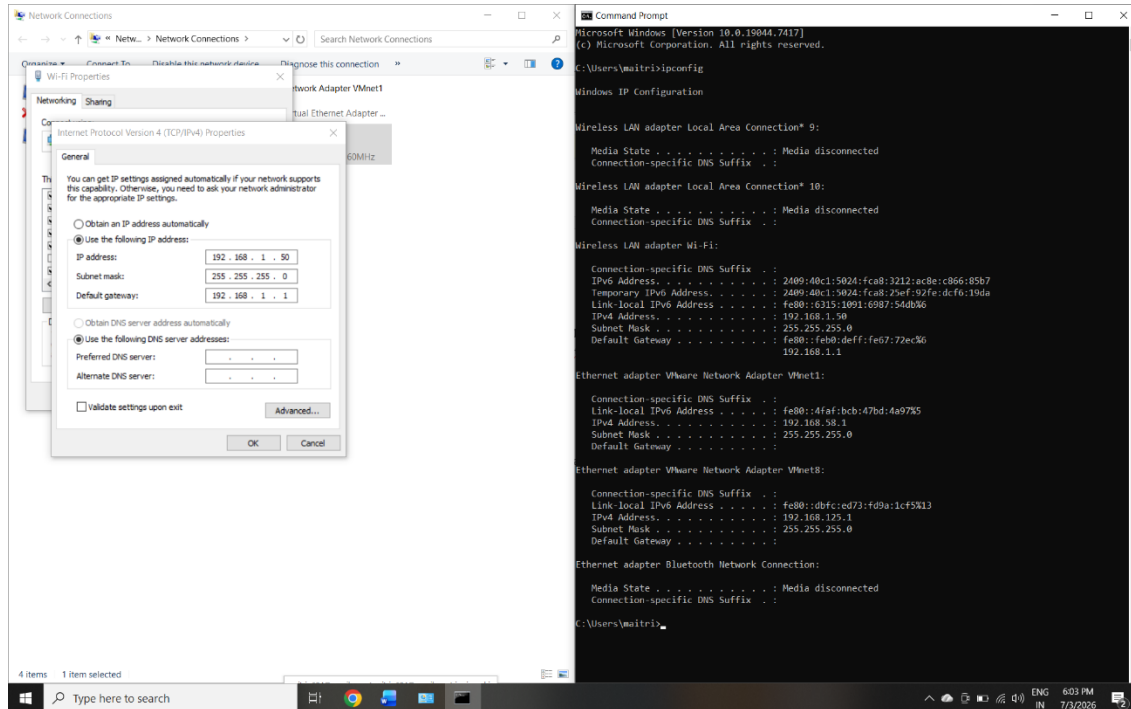
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2401:4900:8f72:339b:d181:cc84:4a9e:5770
    Temporary IPv6 Address. . . . . : 2401:4900:8f72:339b:d81e:6f58:5f10:381c
    Link-local IPv6 Address . . . . . : fe80::6315:1091:6987:54db%6
    IPv4 Address. . . . . : 192.168.1.18
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::a291:caff:fe28:3381%6
                                192.168.1.1

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::4faf:bcdb:47bd:4a97%5
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3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access. Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.



4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

If two devices are assigned the same IP address, an IP address conflict occurs. As a result, one or both devices may lose internet access, experience network connectivity problems, or be unable to communicate properly with other devices on the network.